

CSE563 — Assignment 3

Unit III: Probability and Statistics

Maps to CO3 (apply the concepts of probability and statistics for inference and regression analysis).
Marks: 50. **Attempt all questions. Standard normal table values are supplied where needed.**

Standard normal values supplied for this paper								
z	0.5	1.0	1.28	1.5	1.645	1.96	2.0	2.58
$\Phi(z)$	0.6915	0.8413	0.9000	0.9332	0.9500	0.9750	0.9772	0.9950
$t_{0.025,9} = 2.262, \quad t_{0.025,15} = 2.131, \quad t_{0.025,3} = 3.182$								

Part A — Short Answer and Definitions (10 marks, 2 each)

Q1. [2 marks] Distinguish between mutually exclusive and independent events. Can two events with non-zero probability be both?

Q2. [2 marks] State the Central Limit Theorem and explain what “standard error” means. If a sample of 100 has $\sigma = 20$, what is the standard error? How large a sample would halve it?

Q3. [2 marks] A 95% confidence interval for a mean is $[12.4, 15.6]$. State precisely what this means, and state one common misinterpretation.

Q4. [2 marks] Define Type I and Type II errors. In a cancer-screening test, which is more serious, and how would you adjust α ?

Q5. [2 marks] Explain the difference between MLE and MAP estimation. Under what condition do they coincide?

Part B — Probability and Distributions (20 marks)

Q6. [5 marks] An IT department buys laptops from three vendors: 50% from A, 30% from B, 20% from C. The failure rates within warranty are 2%, 4% and 5% respectively. (a) What is the probability a randomly chosen laptop fails within warranty? (b) Given that a laptop has failed, what is the probability it came from vendor C? (c) Which vendor is the most likely source of a failed laptop?

Q7. [5 marks] A web server receives requests at an average rate of 4 per second. (a) Name the appropriate distribution and state its mean and variance. (b) Find $P(\text{exactly 2 requests in a second})$. (c) Find $P(\text{at least 2 requests in a second})$. (d) What is the probability of no requests in a 0.5-second window? Use $e^{-4} = 0.0183$, $e^{-2} = 0.1353$.

Q8. [5 marks] API response times are normally distributed with $\mu = 250$ ms and $\sigma = 40$ ms. (a) What fraction of requests exceed 300 ms? (b) What fraction fall between 210 ms and 290 ms? (c) The team wants to publish a “p90” latency figure (the value exceeded by only 10% of requests). Compute it. (d) If the SLA is “95% of requests under 320 ms”, is it met?

Q9. [5 marks] A new build process is claimed to have mean compile time 10 minutes. A sample of $n = 16$ builds gives $\bar{x} = 11.2$ min with $s = 2.4$ min. (a) State H_0 and H_1 for a two-sided test. (b) Compute the test statistic and state the degrees of freedom. (c) Test at $\alpha = 0.05$ using $t_{0.025,15} = 2.131$. (d) Construct the 95% confidence interval and confirm it agrees with your decision.

Part C — Bayesian Inference and Regression (20 marks)

Q10. [6 marks] A spam filter is being evaluated. Of 12 emails tested from a new sender, 9 were spam. (a) Give the MLE of the spam probability θ . (b) Using a Beta(2, 2) prior, derive the posterior distribution and state its mean and MAP estimate. (c) Repeat with a Beta(1, 1) prior and comment on the relationship to the MLE. (d) Explain in one sentence what would happen to the influence of the prior if 1200 emails were observed with the same 75% spam rate.

Q11. [8 marks] The table gives hours of CPU profiling effort x and the resulting percentage speed-up y for six modules:

x	2	4	6	8	10	12
y	5	9	14	17	22	27

- (a) Compute $\bar{x}$, $\bar{y}$, S_{xx} , S_{yy} and S_{xy} . (b) Find the least-squares regression line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$. (c) Compute the correlation coefficient r and the coefficient of determination R^2 ; interpret R^2 in one sentence. (d) Predict the speed-up for 9 hours of effort. (e) State one reason why predicting at $x = 40$ would be unwise.

Q12. [6 marks] Using the regression from the previous question: (a) Compute the fitted values and residuals for $x = 2$, $x = 6$ and $x = 12$. (b) Compute SSE and SST, and verify $R^2 = 1 - \text{SSE}/\text{SST}$. (c) Test whether the slope is significantly different from zero at $\alpha = 0.05$, given $t_{0.025,4} = 2.776$ and

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\text{SSE}/(n-2)}{S_{xx}}}.$$